Regularization

Regularization

A fundamental problem is that the algorithm tries to pick parameters that minimize loss on the training set, but this may not result in a model that has low loss on future data. This is called **overfitting**.

example

suppose we want to predict the probability of heads when tossing a coin. We toss it $N = 3$ times and observe 3 heads. The MLE is $\theta_{\text{mle}} = N_1/(N_0 + N1) = 3/(3 + 0) = 1$. However, if we use $\text{Ber}\left(y \mid \theta_{\text{mle}}\right)$ as our model, we will predict that all future coin tosses will also be heads, which seems rather unlikely.

Regularization

- The core of the problem is that the model has enough parameters to perfectly fit the observed training data. so it can **perfectly match** the empirical distribution.
- However, in most cases the empirical distribution is not the same as the true distribution, so putting all the probability mass on the observed set of N examples will not leave over any probability for novel data in the future. That is, **the model may not generalize**.

Solution

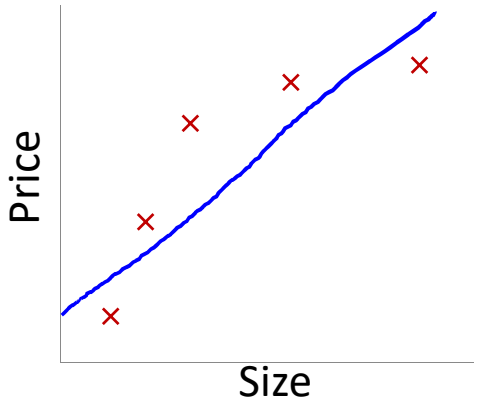
The main solution to overfitting is to use regularization, which means to add a penalty term to the Cost function. Thus we optimize an objective of the form

$$\mathcal{L}(\boldsymbol{\theta}; \lambda) = \left[\frac{1}{M}\sum_{i=1} \ell(\mathbf{y}_i, \boldsymbol{\theta}; \mathbf{x}_i)\right] + \lambda C(\boldsymbol{\theta})$$

$\lambda \geq 0$ is a tuning parameter and control the relative impact of these two terms on the regression coefficient estimates.

When $\lambda = 0$, the penalty term has no effect
However, as $\lambda \to \infty$, the impact of the shrinkage penalty grows, and the ridge regression coefficient estimates will approach zero.

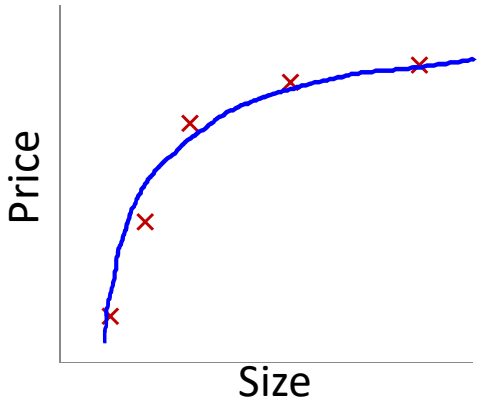
Example: Linear regression (housing prices)



$\rightarrow \theta_0 + \theta_1 x$

"Underfit" "High bias"

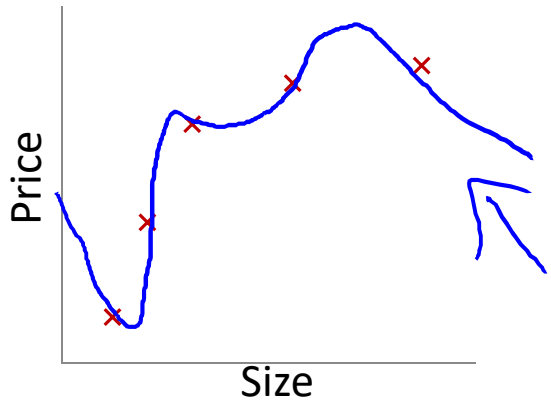
λ ↑



$\rightarrow \theta_0 + \theta_1 x + \theta_2 x^2$

"Just right"

λ



$\rightarrow \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

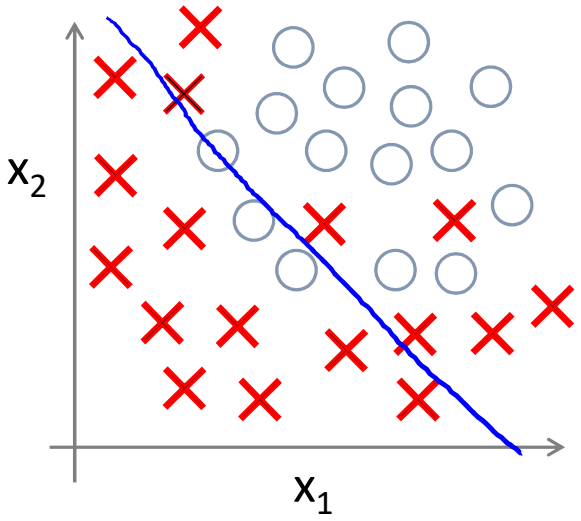
"Overfit" "High variance"

$\lambda = 0$

Overfitting: If we have too many features, the learned hypothesis may fit the training set very well ($J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2 \approx 0$), but fail to generalize to new examples (predict prices on new examples).

Example: Logistic regression

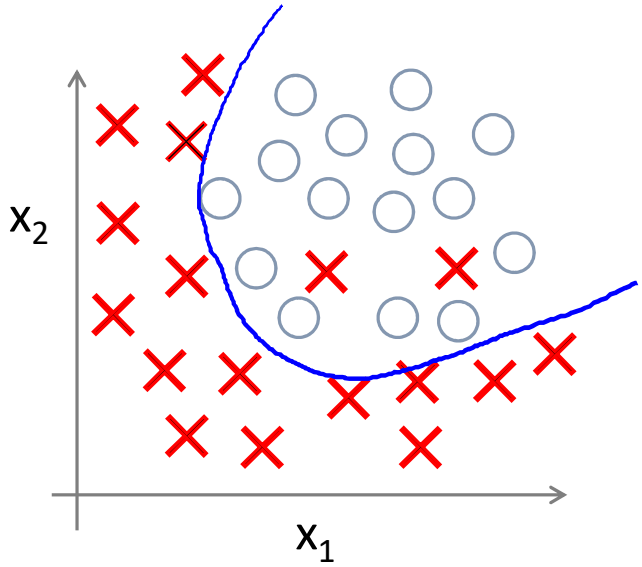
(طبقه‌بندی)



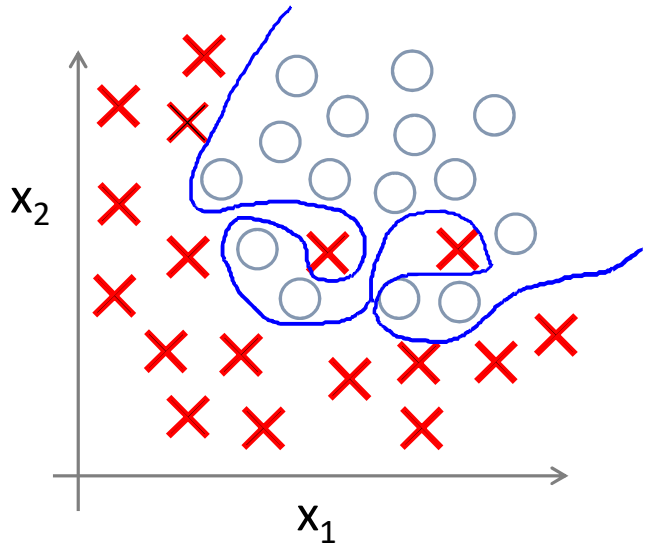
$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$

(g = sigmoid function)

"Underfit"



$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$

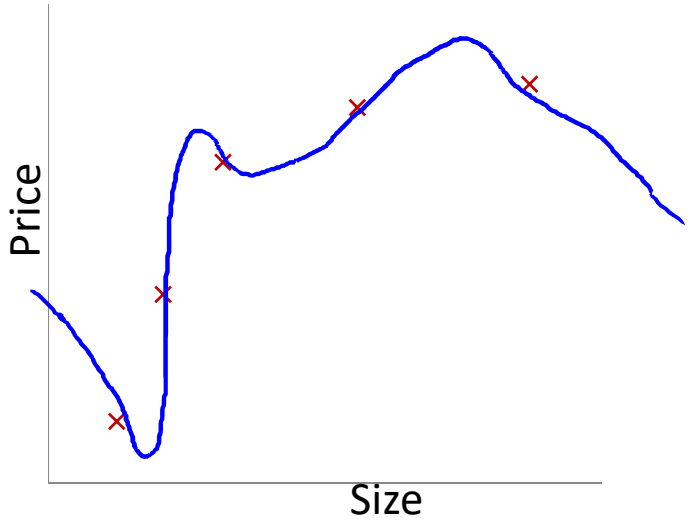


$g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \dots)$

"Overfit"

Addressing overfitting:

$x_1 =$ size of house
$x_2 =$ no. of bedrooms
$x_3 =$ no. of floors
$x_4 =$ age of house
$x_5 =$ average income in neighborhood
$x_6 =$ kitchen size
$\vdots$
x_{100}



Addressing overfitting:

Options:

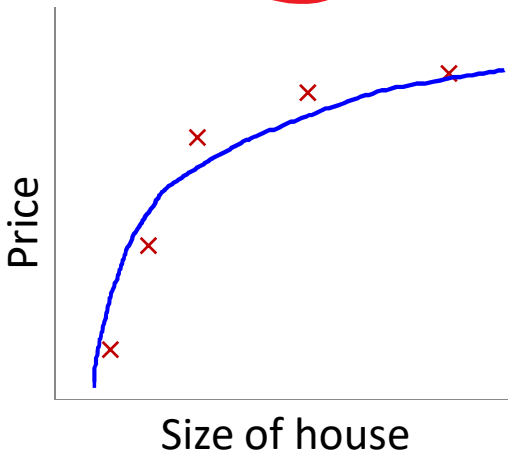
1. Reduce number of features
   - Manually select which features to keep.
   - Model selection algorithm (later in course).
2. Regularization
   - Keep all the features, but reduce magnitude/values of parameters θ_j
   - Works well when we have a lot of features, each of which contributes a bit to predicting y.

Regularization

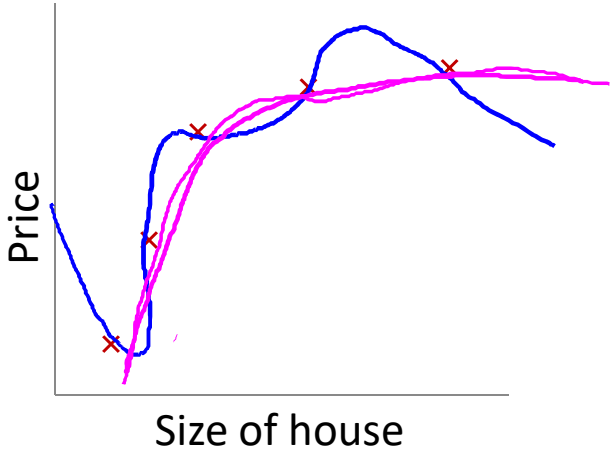
Cost function

Intuition

Bias-Variance tradeoff



$\theta_0 + \theta_1 x + \theta_2 x^2$



$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

Suppose we penalize and make θ_3, θ_4 really small.

$$\min_\theta \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2 + 1000\,\theta_3^2 + 1000\,\theta_4^2$$

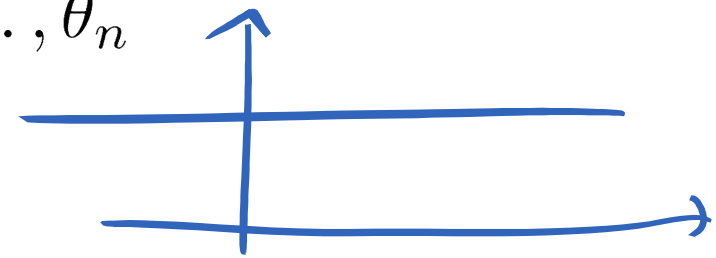
$\theta_3 \approx 0 \qquad \theta_4 \approx 0$

Regularization.

$y = \theta_0 \leftarrow \theta_1 = \theta_2 = \cdots \theta_{100} = 0$

Small values for parameters $\theta_0, \theta_1, \ldots, \theta_n$

- "Simpler" hypothesis
- Less prone to overfitting



Housing:

- Features: $x_1, x_2, \ldots, x_{100}$
- Parameters: $\theta_0, \theta_1, \theta_2, \ldots, \theta_{100}$

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n} \theta_j^2\right]$$

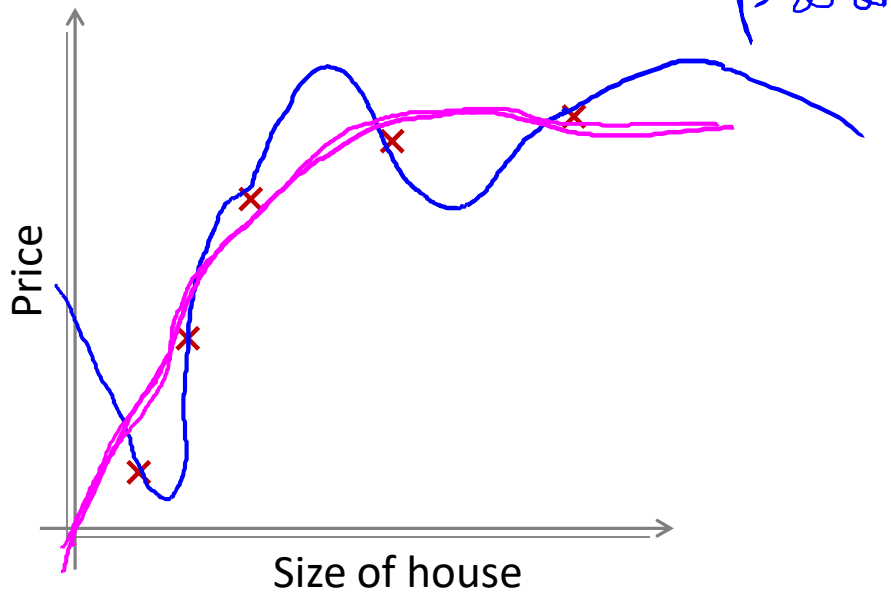
$\theta_1, \theta_2, \theta_3, \ldots, \theta_{100}$

Regularization.

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n}\theta_j^2\right]$$

regularization parameter

$$\min_\theta J(\theta)$$



L1_norm: $\lambda \sum_{j=1}^{n} |\theta_j|$

In regularized linear regression, we choose θ to minimize

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n} \theta_j^2\right]$$

L2_norm regularization

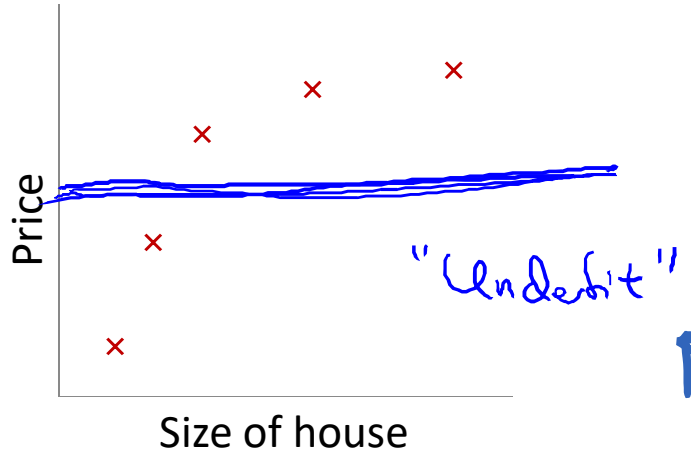
What if λ is set to an extremely large value (perhaps for too large for our problem, say $\lambda = 10^{10}$)?

- Algorithm works fine; setting λ to be very large can't hurt it
- Algortihm fails to eliminate overfitting.
- Algorithm results in underfitting. (Fails to fit even training data well).
- Gradient descent will fail to converge.

In regularized linear regression, we choose θ to minimize

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda\sum_{j=1}^{n}\theta_j^2\right]$$

What if λ is set to an extremely large value (perhaps for too large for our problem, say $\lambda = 10^{10}$)?



$h_\theta(x)$

$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

$\theta_1, \theta_2, \theta_3, \theta_4$

$\theta_1 \approx 0, \theta_2 \approx 0$

$\theta_3 \approx 0, \theta_4 \approx 0$

$h_\theta(x) = \theta_0$

Regularization

Regularized linear regression

Regularized linear regression

$$J(\theta) = \frac{1}{2m}\left[\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^{n} \theta_j^2\right]$$

$$\min_\theta J(\theta)$$

Gradient descent

l_2-norm

Repeat $\{$

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\theta_j := \theta_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)} - \frac{\lambda}{m} \theta_j \right]$$

$$(j = 1, 2, 3, \ldots, n)$$

$\frac{\partial}{\partial \theta_j} J(\theta)$

$\}$

$$\theta_j := \theta_j \left(1 - \alpha \frac{\lambda}{m}\right) - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$1 - \alpha \frac{\lambda}{m} < 1$

0.99

$\theta_j \times 0.99$

Normal equation(Regularized) (L_2-norm)

$$X = \begin{bmatrix} (x^{(1)})^T \\ \vdots \\ (x^{(m)})^T \end{bmatrix} \qquad y = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix} \in \mathbb{R}^m$$

$m \times (n+1)$

$$\Rightarrow \min_\theta J(\theta)$$

$$\frac{\partial}{\partial \theta_j} J(\theta) \overset{\text{set}}{=} 0$$

$$\theta = \left(X^T X + \lambda \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix} \right)^{-1} X^T y$$

$(n+1) \times (n+1)$

e.g. $n = 2$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Non-invertibility

Suppose $m \leq n$,

(#examples) (#features)

$$\theta = (X^T X)^{-1} X^T y$$

non-invertible / singular

If $\lambda > 0$,

$$\theta = \left(X^T X + \lambda \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix} \right)^{-1} X^T y$$

invertible

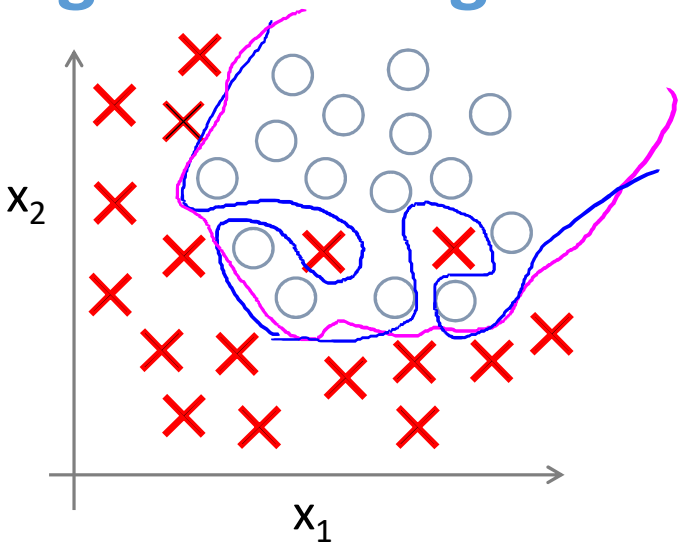
$m \ll n$

ویژگی‌های وابسته خطی

Regularization

Regularized logistic regression

Regularized logistic regression.



$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \dots)$$

Cost function:

$$J(\theta) = -\left[\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)}))\right]$$

Reg. Term

$+ \frac{\lambda}{2m} \sum_{j=1}^{n} \theta_j^2$

$\theta_1, \theta_2, \dots, \theta_n$

Gradient descent

$h_\theta(x^{(i)}) = g(\theta^T x^{(i)})$

$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2$

$x_3 \quad x_4$

Repeat $\{$

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_0^{(i)}$$

$$\theta_j := \theta_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)} - \frac{\lambda}{m} \theta_j \right]$$

$(j = \cancel{0}, 1, 2, 3, \ldots, n)$

$\theta_1 \ldots \theta_n$

$\frac{\partial}{\partial \theta_j} J(\theta)$

$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$

$\}$

$$J(\theta) = \dots + \frac{\lambda}{2m} \sum_{j=1} |\theta_j|$$

L1_norm

Sparsity ← L0_norm

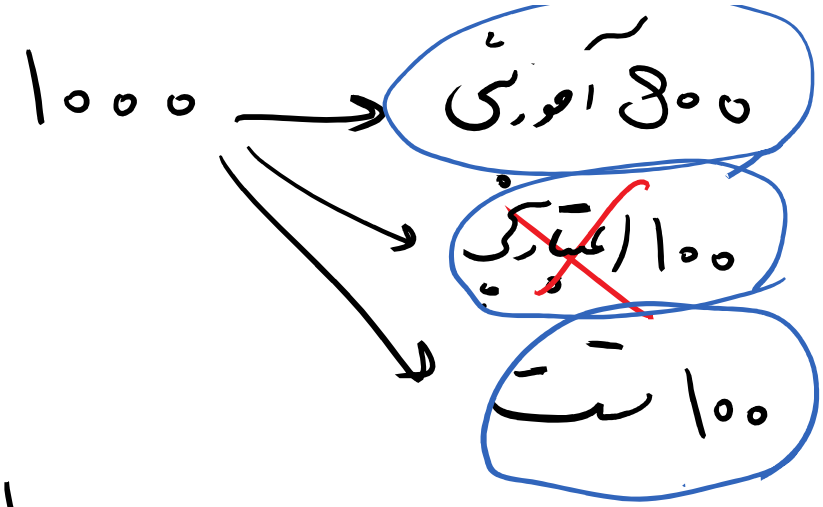
Bayesian approach ⟶ به جای تخمین نقطه‌ای برای θ، توزیع احتمال θ تخمین زده می‌شود و عدم قطعیت هم قابل می‌شود.

$p(\theta)$ θ تصادفی با توزیع مشخص

θ را بردار Sparsity تخمین بزنید.

$$\hat{\theta}_{MAP} = \max_{\theta} p(\theta | x)$$

MAP

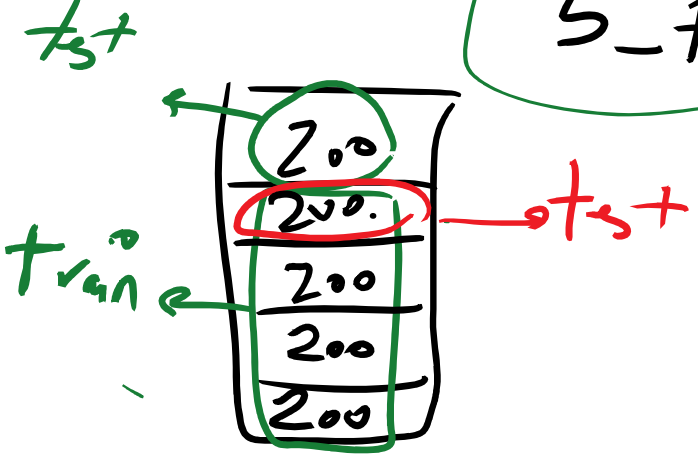


معیار ارزیابی -

۱۰۰۰

5_fold

(K - Fold cross Validation)



Probabilistic Learning

Probabilistic interpretation of Linear Regression

$y^{(i)} = \theta^T X^{(i)} + \epsilon^{(i)}$ i = 1 … m

$\epsilon^{(i)}$: Gaussian , iid

$\epsilon^{(i)} \sim N(0, \sigma^2)$ $i = 1 \dots m$

$$p\left(y^{(i)}|x^{(i)}, \theta\right) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(y^i - \theta^\top x^{(i)}\right)^2}{2\sigma^2}\right)$$

$\underline{y} = [y^{(1)}, y^{(2)}, \dots y^{(m)}]^T$

$l(\theta) = P_\theta(\underline{y}|x) = \prod_{i=1}^m p(y^{(i)}|x^{(i)}, \theta)$

Likelihood function

$L(\theta) = \log l(\theta)$

log_likelihood

$\max_\theta L(\theta) = \min \frac{1}{2}\sum_{i=1}^m (y^{(i)} - \theta^T x^{(i)})^2$

معادل $l(\theta)$ در رگرسیون خطی

Least squares

$\max_\theta - \sum_{i=1}^m (y^{(i)} - \theta^T x^{(i)})^2$

(که نسبتی در تخمین θ ندارد.)

Probabilistic interpretation of logistic regression

$$h_\theta(x) = p(y = 1|x, \theta) \qquad 1 - h_\theta(x) = p(y = 0|x, \theta)$$

$\sigma(\theta^T x)$

$$p\left(y^{(i)}|x^{(i)}, \theta\right) = \left(h_\theta\left(x^{(i)}\right)\right)^{y^{(i)}} \left(1 - h_\theta\left(x^{(i)}\right)\right)^{1-y^{(i)}}$$

$Ber(y;p) = p^y (1-p)^{1-y}$

$$l(\theta) = \prod_{i=1}^{m} p\left(y^{(i)}|x^{(i)}, \theta\right) = \prod_{i=1}^{m} \left(\left(h_\theta\left(x^{(i)}\right)\right)^{y^{(i)}} \left(1 - h_\theta\left(x^{(i)}\right)\right)^{1-y^{(i)}}\right)$$

$$L(\theta) = \log L(\theta) = \sum_{i=1}^{m} y^{(i)} \log h_\theta\left(x^{(i)}\right) + \left(1 - y^{(i)}\right) \log\left(1 - h_\theta\left(x^{(i)}\right)\right) = -J(\theta)$$

log_likelihood

cross_entropy

ML حل

$$\max_\theta L(\theta) = \min_\theta -L(\theta) = \min_\theta J(\theta)$$

Generalized Linear Models

GLM

In the regression example and the classification we had:

$$y \mid x; \theta \sim \mathcal{N}(\mu, \sigma^2)$$

$$y \mid x; \theta \sim \text{Bernoulli}(\phi)$$

In this section, we will show that both of these methods are special cases of a broader family of models, called Generalized Linear Models (GLMs).

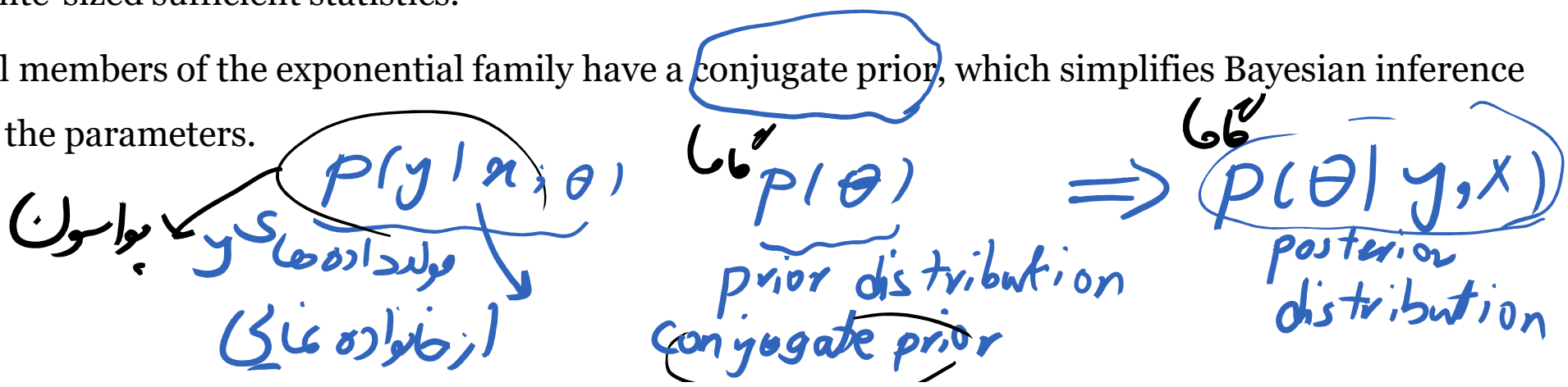
We will also show how other models in the GLM family can be derived and applied to other classification and regression problems.

$p(y \mid x)$

The exponential family

The exponential family plays a crucial role in statistics and machine learning, for various reasons, including the following:

- The exponential family is the unique family of distributions that has maximum entropy.
- The exponential family is at the core of GLMs.
- The exponential family is at the core of variational inference.
- Under certain regularity conditions, the exponential family is the only family of distributions with finite-sized sufficient statistics.
- All members of the exponential family have a conjugate prior, which simplifies Bayesian inference of the parameters.



The exponential family

We say that a class of distributions is in the exponential family if it can be written in the form:

$$p(y;\eta) = b(y)\exp\left(\eta^T T(y) - a(\eta)\right) \quad \textbf{(1)}$$

η: natural parameter

$T(y)$:sufficient statistic

$a(\eta)$: log partition function

پارامتر طبیعی

The quantity $e^{-a(\eta)}$ essentially plays the role of a normalization constant, that makes sure the distribution $p(y;\eta)$ sums/integrates over y to 1 .

A fixed choice of T, a and b defines a family (or set) of distributions that is parameterized by η; as we vary η, we then get different distributions within this family.

Bernoulli distribution

The Bernoulli distribution with mean ϕ, written Bernoulli (ϕ), specifies a distribution over $y \in \{0,1\}$, so that $p(y = 1; \phi) = \phi; p(y = 0; \phi) = 1 - \phi$.

there is a choice of T, a and b so that Equation (1) becomes exactly the class of Bernoulli distributions.

Bernoulli distribution

We write the Bernoulli distribution as:

$$\begin{aligned} p(y;\phi) &= \phi^{y}(1-\phi)^{1-y} \\ &= \exp(y\log\phi + (1-y)\log(1-\phi)) \\ &= \exp\left(\left(\log\left(\frac{\phi}{1-\phi}\right)\right)y + \log(1-\phi)\right) \end{aligned}$$

$\phi = e^{\log\phi}$

Thus, the natural parameter is given by $\eta = \log(\phi/(1-\phi))$.

$T(y) = y$

if we invert this definition for η by solving for ϕ in terms of η, we obtain $\phi = 1/(1+e^{-\eta})$.

Bernoulli distribution

To complete the formulation of the Bernoulli distribution as an exponential family distribution, we also have:

$$
\begin{aligned}
T(y) &= y \\
a(\eta) &= -\log(1-\phi) \\
&= \log(1+e^{\eta}) \\
b(y) &= 1
\end{aligned}
$$

This shows that the Bernoulli distribution can be written in the form of Equation (1), using an appropriate choice of T, a and b.

Gaussian distribution

Recall that, when deriving linear regression, the value of σ^2 had no effect on our final choice of θ and $h_\theta(x)$. Thus, we can choose an arbitrary value for σ^2 without changing anything. To simplify the derivation below, let's set $\sigma^2 = 1$. We then have:

$$p(y;\mu) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}(y-\mu)^2\right)$$
$$= \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}y^2\right)\cdot\exp\left(\mu y - \frac{1}{2}\mu^2\right)$$

چون پارامتر $\sigma^2 = 1$ در رابطه صدق می‌کند

در حالی که $\eta = \sigma^2 \neq 1$ و پارامتر طبیعی می‌شود

$\eta = \mu \quad T(y) = y \quad a(\eta) = \frac{\mu^2}{2} = \frac{\eta^2}{2}$

$b(y) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y^2}{2}\right)$

Gaussian distribution

we see that the Gaussian is in the exponential family, with:

$$
\begin{aligned}
\eta &= \mu \\
T(y) &= y \\
a(\eta) &= \mu^2/2 \\
&= \eta^2/2 \\
b(y) &= (1/\sqrt{2\pi})\exp\left(-y^2/2\right)
\end{aligned}
$$

توزیع های Categorical ، پواسون (برای داده‌های شمارشی) ، ماتاوه‌نمایی
(مناسب برای داده پیوسته و ناصفی مثل بازه زمانی) ، بتا ، دیریکله

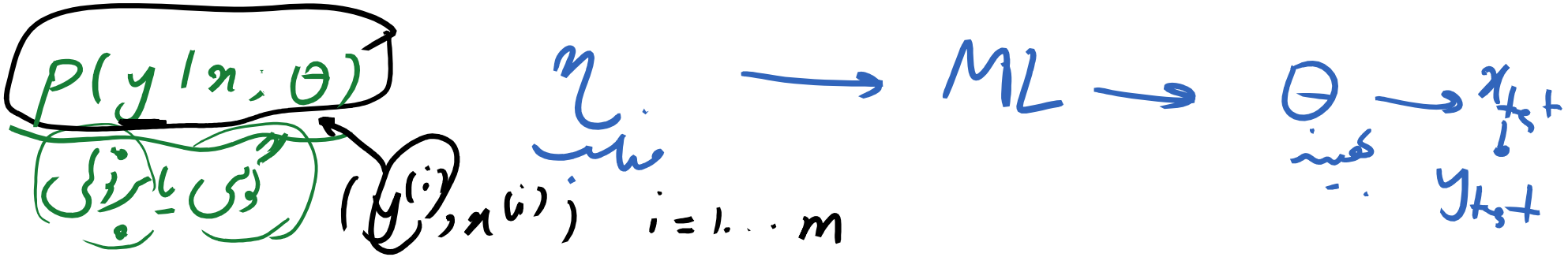
(مناسب برای توزیع احتمال یک بار تصادفی مثلاً توزیع وی ϕ در رولی



همگی از خانواده نمایی هستند و در کاربرد های مختلف می‌توانند مفید واقع شوند

Constructing GLMs

- Suppose you would like to build a model to estimate the number y of customers arriving in your store (or number of page-views on your website) in any given hour, based on certain features x such as store promotions, recent advertising, weather, day-of-week, etc.
- We know that the **Poisson** distribution usually gives a good model for numbers of visitors.
- Poisson is an exponential family distribution, so we can apply a Generalized Linear Model (GLM).



Constructing GLMs

- More generally, consider a classification or regression problem where we would like to predict the value of some random variable y as a function of x.
- To derive a GLM for this problem, we will make the following three assumptions about the conditional distribution of y given x and about our model:

1. $y \mid x; \theta \sim$ Exponential Family (η). I.e., given x and θ, the distribution of y follows some exponential family distribution, with parameter η.
2. Given x, our goal is to predict the expected value of $T(y)$ given x. In most of our examples, we will have $T(y) = y$, so this means we would like the prediction $h(x)$ output by our learned hypothesis h to satisfy $h(x) = \mathrm{E}[y \mid x]$.
3. The natural parameter η and the inputs x are related linearly: $\eta = \theta^T x$. (Or, if η is vector-valued, then $\eta_i = \theta_i^T x$.)

$h(x) = E[T(y) \mid x]$

Ordinary Least Squares

- To show that ordinary least squares is a special case of the GLM family of models, consider the setting where the target variable y (also called the response variable in GLM terminology) is continuous, and we model the conditional distribution of y given x as a Gaussian $\mathcal{N}(\mu, \sigma^2)$. (Here, μ may depend x.)
- let the Exponential Family distribution above be the Gaussian distribution.
- As we saw previously, in the formulation of the Gaussian as an exponential family distribution, we had $\mu = \eta$.
- So, we have:

$$\begin{aligned} h_\theta(x) &= E[y \mid x; \theta] \\ &= \mu \\ &= \eta \\ &= \theta^T x. \end{aligned}$$

$h(x) = E[T(y) \mid x] = \theta^T x$

Logistic Regression

In our formulation of the Bernoulli distribution as an exponential family distribution, we had:

$\phi = 1/(1 + e^{-\eta})$.

if $y \mid x; \theta \sim \text{Bernoulli}(\phi)$, then $\text{E}[y \mid x; \theta] = \phi$. So, following a similar derivation as the one for ordinary least squares, we get:

$$\begin{aligned} h_\theta(x) &= E[y \mid x; \theta] \\ &= \phi \\ &= 1/(1 + e^{-\eta}) \\ &= 1/\left(1 + e^{-\theta^T x}\right) \end{aligned}$$

So, this gives us hypothesis functions of the form $h_\theta(x) = 1/\left(1 + e^{-\theta^T x}\right)$

Softmax Regression

- Consider a classification problem in which the response variable y can take on any one of k values, so $y \in \{1,2, \ldots, k\}$.
- For example, rather than classifying email into the two classes spam or not-spam-which would have been a binary classification problem might want to classify it into three classes, such as spam, personal mail, and work-related mail.
- The response variable is still discrete, but can now take on more than two values. We will thus model it as distributed according to a categorical distribution.

برای مسئله k طبقه

$\text{Categorical}(\phi_1, \phi_2, \ldots, \phi_{k-1})$ $\quad \phi_k = 1 - \sum_{i=1}^{k-1} \phi_i$

Softmax Regression

- To parameterize a multinomial over k possible outcomes, one could use k parameters $\phi_1, \ldots, \phi_k$ specifying the probability of each of the outcomes. However, these parameters would be redundant, or more formally, they would not be independent ($\sum_{i=1}^{k} \phi_i = 1$).
- So, we will instead parameterize the categorical with only $k-1$ parameters, $\phi_1, \ldots, \phi_{k-1}$, where $\phi_i = p(y = i; \phi)$, and $p(y = k; \phi) = 1 - \sum_{i=1}^{k-1} \phi_i$.
- For notational convenience, we will also let $\phi_k = 1 - \sum_{i=1}^{k-1} \phi_i$, but we should keep in mind that this is not a parameter, and that it is fully specified by $\phi_1, \ldots, \phi_{k-1}$.

Softmax Regression

- To express the categorical as an exponential family distribution, we will define $T(y) \in \mathbb{R}^{k-1}$ as follows:

$$T(1) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, T(2) = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, T(3) = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, T(k-1) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}, T(k) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

$$(T(y))_i = 1\{y = i\}$$

Softmax Regression

We are now ready to show that the categorical is a member of the exponential family. We have:

$$
\begin{aligned}
p(y;\phi) &= \phi_1^{1\{y=1\}}\phi_2^{1\{y=2\}}\cdots\phi_k^{1\{y=k\}} \\
&= \phi_1^{1\{y=1\}}\phi_2^{1\{y=2\}}\cdots\phi_k^{1-\sum_{i=1}^{k-1}1\{y=i\}} \\
&= \phi_1^{(T(y))_1}\phi_2^{(T(y))_2}\cdots\phi_k^{1-\sum_{i=1}^{k-1}(T(y))_i} \\
&= \exp((T(y))_1\log(\phi_1) + (T(y))_2\log(\phi_2) + \\
&\qquad \cdots + \left(1-\sum_{i=1}^{k-1}(T(y))_i\right)\log(\phi_k)\Bigg) \\
&= \exp((T(y))_1\log(\phi_1/\phi_k) + (T(y))_2\log(\phi_2/\phi_k) + \\
&\qquad \cdots + (T(y))_{k-1}\log(\phi_{k-1}/\phi_k) + \log(\phi_k)\big) \\
&= b(y)\exp\big(\eta^T T(y) - a(\eta)\big)
\end{aligned}
$$

Softmax Regression

where

$$\eta = \begin{bmatrix} \log(\phi_1/\phi_k) \\ \log(\phi_2/\phi_k) \\ \vdots \\ \log(\phi_{k-1}/\phi_k) \end{bmatrix},$$

$$a(\eta) = -\log(\phi_k)$$

$$b(y) = 1.$$

This completes our formulation of the categorical as an exponential family distribution.

The link function is given (for $i = 1, \ldots, k$) by

$$\eta_i = \log \frac{\phi_i}{\phi_k}$$

Softmax Regression

For convenience, we have also defined $\eta_k = \log(\phi_k/\phi_k) = 0$. To invert the link function and derive the response function, we therefore have that

$$e^{\eta_i} = \frac{\phi_i}{\phi_k} \quad (2)$$

$$\phi_k e^{\eta_i} = \phi_i$$

$$\phi_k \sum_{i=1}^{k} e^{\eta_i} = \sum_{i=1}^{k} \phi_i = 1$$

This implies that $\phi_k = 1/\sum_{i=1}^{k} e^{\eta_i}$, which can be substituted back into Equation(2) to give the response function:

$$\phi_i = \frac{e^{\eta_i}}{\sum_{j=1}^{k} e^{\eta_j}}$$

Softmax

$\theta_1, \ldots, \theta_{k-1} \in \mathbb{R}^{n+1}$

$\eta_i = \theta_i^T x \quad (i = 1 \ldots k-1)$

Softmax Regression

This function mapping from the η 's to the ϕ 's is called the softmax function. To complete our model, we use Assumption 3, given earlier, that the η_i 's are linearly related to the x 's. So, have $\eta_i = \theta_i^T x$ (for $i = 1, \ldots, k-1$), where $\theta_1, \ldots, \theta_{k-1} \in \mathbb{R}^{n+1}$ are the parameters of our model.

For notational convenience, we can also define $\theta_k = 0$, so that $\eta_k = \theta_k^T x = 0$, as given previously. Hence, our model assumes that the conditional distribution of y given x is given by

$$\begin{aligned} p(y = i \mid x; \theta) &= \phi_i \\ &= \frac{e^{\eta_i}}{\sum_{j=1}^{k} e^{\eta_j}} \\ &= \frac{e^{\theta_i^T x}}{\sum_{j=1}^{k} e^{\theta_j^T x}} \end{aligned} \tag{3}$$

Softmax Regression

This model, which applies to classification problems where $y \in \{1, \ldots, k\}$, is called softmax regression.

It is a generalization of logistic regression.

Our hypothesis will output:

$$
\begin{aligned}
h_\theta(x) &= \mathrm{E}[T(y) \mid x; \theta] \\
&= \mathrm{E}\begin{bmatrix} 1\{y = 1\} \\ 1\{y = 2\} \\ \vdots \\ 1\{y = k-1\} \end{bmatrix} x; \theta \Bigg] \\
&= \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{k-1} \end{bmatrix} \\
&= \begin{bmatrix} \frac{\exp(\theta_1^T x)}{\sum_{j=1}^{k} \exp\left(\theta_j^T x\right)} \\ \frac{\exp(\theta_2^T x)}{\sum_{j=1}^{k} \exp\left(\theta_j^T x\right)} \\ \vdots \\ \frac{\exp(\theta_{k-1}^T x)}{\sum_{i=1}^{k} \exp\left(\theta_i^T x\right)} \end{bmatrix}
\end{aligned}
$$

$$p(y = k \mid x; \theta) = 1 - \sum_{i=1}^{k-1} \phi_i$$

Softmax Regression

if we have a training set of m examples $\{(x^{(i)}, y^{(i)}); i = 1, \ldots, m\}$ and would like to learn the parameters θ_i of this model, we would begin by writing down the log-likelihood

$$\ell(\theta) = \sum_{i=1}^{m} \log p(y^{(i)} \mid x^{(i)}; \theta)$$

$$= \sum_{i=1}^{m} \log \prod_{l=1}^{k} \left(\frac{e^{\theta_l^T x^{(i)}}}{\sum_{j=1}^{k} e^{\theta_j^T x^{(i)}}} \right)^{1\{y^{(i)}=l\}}$$

To obtain the second line above, we used the definition for $p(y \mid x; \theta)$ given in Equation (3). We can now obtain the maximum likelihood estimate of the parameters by maximizing $\ell(\theta)$ in terms of θ, using a method such as gradient ascent or Newton's method.

$p(y \mid x, \theta) = \mathcal{N}(\mu, 1)$

یا

$T(y) = y$ $\quad \eta = \mu \quad$ $\eta = \theta^T x$

MLE $\rightarrow \theta_{opt} \rightarrow x_{test} \rightarrow p(y \mid x_{test}, \theta_{opt})$

$p(\underline{y} \mid \underline{x}, \theta)$

$\underline{y} = [y^{(1)} \cdots y^{(m)}]^T$

Generative Models for Classification

Introduction

- Previous algorithms modeled $p(y \mid x; \theta)$, the conditional distribution of y given x.
- For instance, logistic regression modeled $p(y \mid x; \theta)$ as $h_\theta(x) = g(\theta^T x)$

$p(y \mid x)$

مدل‌های

Discriminative

sigmoid

Why do we need another method, when we have logistic regression?

- When there is substantial separation between the two classes, the parameter estimates for the logistic regression model are surprisingly unstable.
- If the distribution of the predictors X is approximately normal in each of the classes and the sample size is small, then the approaches in this section may be more accurate than logistic regression.
- The methods can be naturally extended to the case of more than two response classes. in other words can fit classes separately.

$p(x \mid y)$ $p(x \mid y=0)$ $p(x \mid y=1)$

Discriminative learning VS generative learning

- Generative models generate the features x for each class c, by sampling from $p(x \mid y = c; \theta)$.
- Discriminative model directly learns the class posterior $p(y \mid x; \theta)$.

Bayes theorem for classification

$$p(y = c \mid \mathbf{x}; \boldsymbol{\theta}) = \frac{p(\mathbf{x} \mid y = c; \boldsymbol{\theta}) p(y = c; \boldsymbol{\theta})}{\sum_{c'} p(\mathbf{x} \mid y = c'; \boldsymbol{\theta}) p(y = c; \boldsymbol{\theta})}$$

The term $p(y = c; \boldsymbol{\theta})$ is the prior over class labels, and the term $p(\mathbf{x} \mid y = c; \boldsymbol{\theta})$ is called the class conditional density for class c.

$$p(y|x) = \frac{p(x|y)\,p(y)}{p(x)}$$

باینری

$y \in \{0, 1\}$

$$p(x) = p(x|y=0)\,p(y=0) + p(x|y=1)\,p(y=1)$$

$$\arg\max_y p(y|x) = \arg\max_y p(x|y)\,p(y)$$

Gaussian discriminant analysis

- Assume that $p(x \mid y)$ is distributed according to a multivariate normal distribution. $[x_1, x_2, \cdots x_n]^T$

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right).$$

$\Sigma_{n \times n}$ ماتریس کوواریانس

$|\Sigma|$: دترمینان Σ

$\mu = E[X] = \int p(x)\, x \, dx$

$\Sigma = E[(X-\mu)(X-\mu)^T]$

$A^H = A$ ماتریس وصیتی $(A^*)^T = A$

H. conjugate Transpose

حقیقی A $\boxed{A^T = A}$

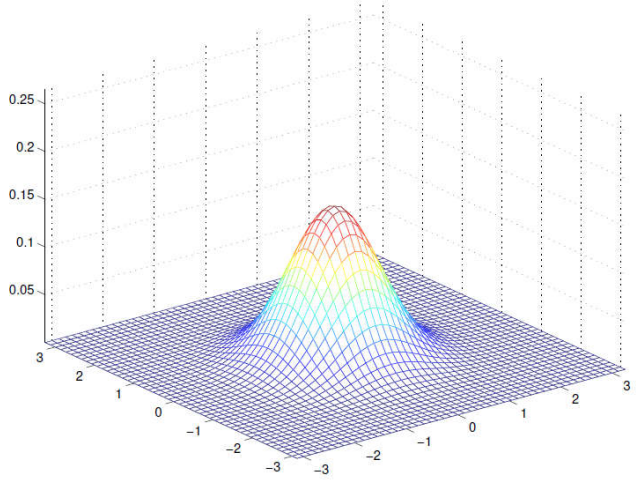
$\forall z$ بردار z $(z^*)^T A z > 0 \Rightarrow$ است A PD

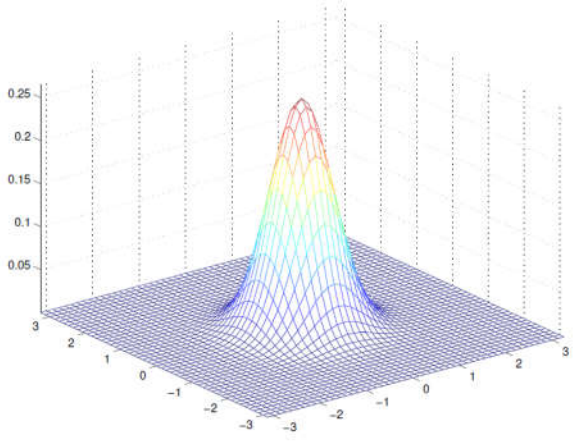
$\geq 0 \Rightarrow$ است PSD A

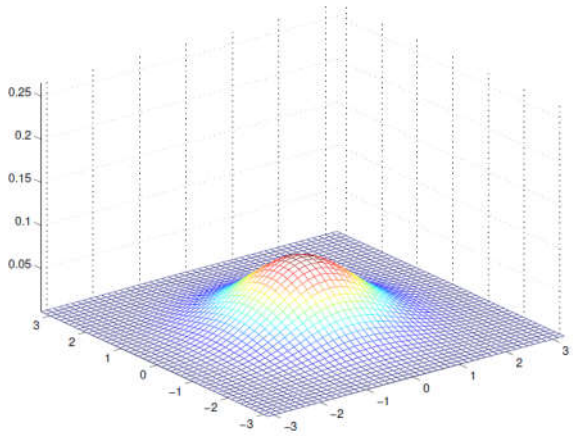
$z z^* = |z|^2$ $A|z|^2 > 0$

$\underline{z}^T \underline{z}^* = (|z_1|^2 \ldots |z_n|^2)$ $A > 0$

Density of a Gaussian distribution



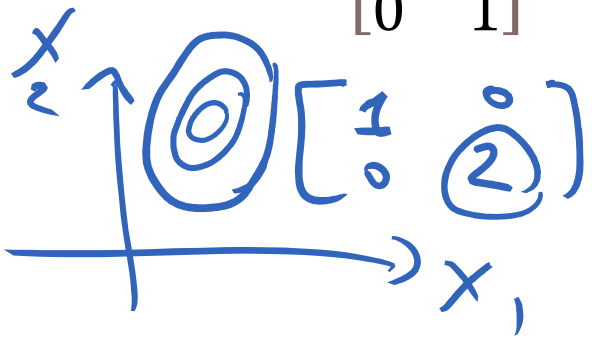




$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$$



$\mu = [0,0]^T$ $\quad \mathcal{N}(0, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix})$ $\quad \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$

The Gaussian Discriminant Analysis model

When we have a classification problem in which the input features x are continuous-valued random variables, we can then use the Gaussian Discriminant Analysis (GDA) model, which models $p(x \mid y)$ using a **multivariate normal distribution**. The model is:

$y \in \{0, 1\}$

$$
\begin{aligned}
y &\sim \text{Bernoulli}(\phi) = \phi^y (1-\phi)^{1-y} \\
x \mid y = 0 &\sim \mathcal{N}(\mu_0, \Sigma) \\
x \mid y = 1 &\sim \mathcal{N}(\mu_1, \Sigma)
\end{aligned}
$$

$$p(y) = \phi^y (1-\phi)^{1-y}$$

$$p(x \mid y=0) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} exp\left(-\frac{1}{2}(x-\mu_0)^T \Sigma^{-1}(x-\mu_0)\right)$$

$$p(x \mid y=1) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} exp\left(-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)\right)$$

Model parameters: $\sum, \mu_1, \mu_2, \phi$

log_likelihood:

$$L(\phi, \mu_0, \mu_1, \Sigma) = \log \prod_{i=1}^{m} p(x^{(i)}, y^{(i)}; \phi, \mu_0, \mu_1, \Sigma)$$

$$l(\phi, \mu_0, \mu_1, \Sigma) = \log \prod_{i=1}^{m} \left[p(x^{(i)} | y^{(i)}, \mu_0, \mu_1, \Sigma) \, p(y^{(i)}; \phi) \right]$$

$$= \sum_{i=1}^{m} \Big(p(x^{(i)} | y^{(i)}, \mu_0, \mu_1, \Sigma) \underbrace{p(y^{(i)}; \phi)}_{\phi^{y^{(i)}}(1-\phi)^{(1-y^{(i)})}} \Big)$$